

Virtual Ultrasonic Parking Sensor

- A college software-engineering capstone project
- Languages: C++17 (app, client, tests) + C (kernel driver)
- Platform: Linux (Ubuntu)
- Submitted via GitHub with full README and docs

What is it?

- Simulates a car parking sensor in software
- Measures a fake distance 150 cm -> 10 cm -> back
- Classifies it: SAFE / WARNING / DANGER
- Logs every reading and serves live status over TCP

System Architecture

- Kernel timer -> /dev/parking_sensor (kernel driver)
- parking_monitor (C++) reads the device
- ThresholdPolicy decides the state
- Logger writes logs/parking.log
- TCP server on 127.0.0.1:9000 -> status_client

Threshold Rules

- distance > 50 cm $\rightarrow$ SAFE
- $21 \text{ cm} \leq \text{distance} \leq 50$ $\rightarrow$ WARNING
- distance ≤ 20 cm $\rightarrow$ DANGER

Key Components

- driver/parking_sensor.c kernel character device driver
- app/ parking_monitor main C++ application
- client/ status_client TCP client (shows live status)
- tests/ 41 tests / 1140 checks, all passing
- docs/ architecture, requirements, test report

Design Decisions

- Driver reports distance only; state logic lives in user space
- --simulate mode swaps in a virtual sensor (no root needed)
- ISensorSource + ThresholdPolicy interfaces = testable design
- No external libraries - only g++, make, POSIX

How to Run

- `make` `# build everything`
- `make test` `# 1140 checks, all pass`
- `./build/parking_monitor --simulate --iterations 20`
- `./build/status_client --port 9000 --count 3`
- Full driver run: `make driver; sudo ./scripts/load_driver.sh`

Testing

- Unit tests: state machine, config parser, logger, telemetry
- Integration test: full monitor run writes all states to log
- TCP test: live server/client check
- Result: 1140 checks run, 0 failed

Conclusion

- Demonstrates: Linux device drivers, file I/O, threads, TCP sockets
- Clean layered architecture, fully tested
- Works with or without root (--simulate)
- GitHub: github.com/rajarp3108-tech/Virtual_Sonic_Parking